Assignment 1: Your first Gemini call

This week, you will:

- Set up your GitHub repo and development environment.
- Learn how LLM APIs work and send your first request to Gemini.
- Experiment with responses and build the foundation for an agent. *We are not building an agent yet*.

Get set up on Slack

- Add a profile picture and create a public project channel named `#p_[name]`, for example `#p_ed_sioufi` (p stands for project).
- Use your channel for progress, questions, and feedback. Everyone is welcome to ask questions and encouraged to help in every project channel.

Your first LLM API call

- Start in a local project folder. Read the Gemini API quickstart to understand requests and responses, then make your first call using Python and google-genai. Get a key using the API key guide.
- Set your key as the `GEMINI_API_KEY`, outside your code. Keep keys and `.env` files out of Git. Choose a model with available free-tier quota; no paid upgrade is needed.
  Tip: Avoid unnecessary repeated calls. If you hit a rate limit, pause and check when your quota resets.
- Adapt the prompt to a cybersecurity example you understand, such as a code snippet you wrote. Start with one API request and display the response.

Experiment and document your work

- Try at least three inputs of your choice. Note your expectations before each run, then assess what Gemini gets right, misses, or claims without evidence. Use your cybersecurity knowledge to check at least one claim.
- Prepare your code, dependencies, and a README with instructions to run it. Record the model, inputs, prompts, responses, and a short reflection.

Share your work on GitHub

- New to Git and pull requests? Use one of these guides as a reference for your own repo: GitHub Desktop (graphical app), Hello World (browser), or pull request quickstart (terminal, GitHub CLI tab). Use whichever workflow you prefer.
  I personally use the GitHub CLI. It's worth exploring, but choose whichever interface helps you get started.
- Create your repo and make your first commit on main: a `README` describing your Gemini experiment, plus a Python `.gitignore`. Then create a new branch and add your code and expand the `README` with setup instructions and results. These changes will be your first PR.
- Invite me using my Github handle @edsioufi as a collaborator. Post your repo link in your Slack project channel and pin the message.
- A pull request (PR) proposes changes for review before they are merged. Your deliverable is your first PR from your branch into main. Request a review from me and share the PR link in your Slack channel before our next workshop. Wait for my review and approval before merging.

Explore further

You can go beyond a terminal script: try a simple interface, read input from a file, or use the response in a small workflow. Choose an extension that interests you. You can also compare prompts or add context.

If you need inspiration or help, tag me or others in a question in your slack channel